

Force & Motion

Use the cue, then fade the cue. Study the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS A **force** is a push or a pull. Forces can start, stop, speed up, slow down, or change the **direction** of an object. **Gravity** pulls objects down. **Friction** opposes motion and is greater on rough surfaces. Motion is measured with **distance** and **time**.

Round 1 — Retrieve It

1. What is a force? Use the words push and pull.

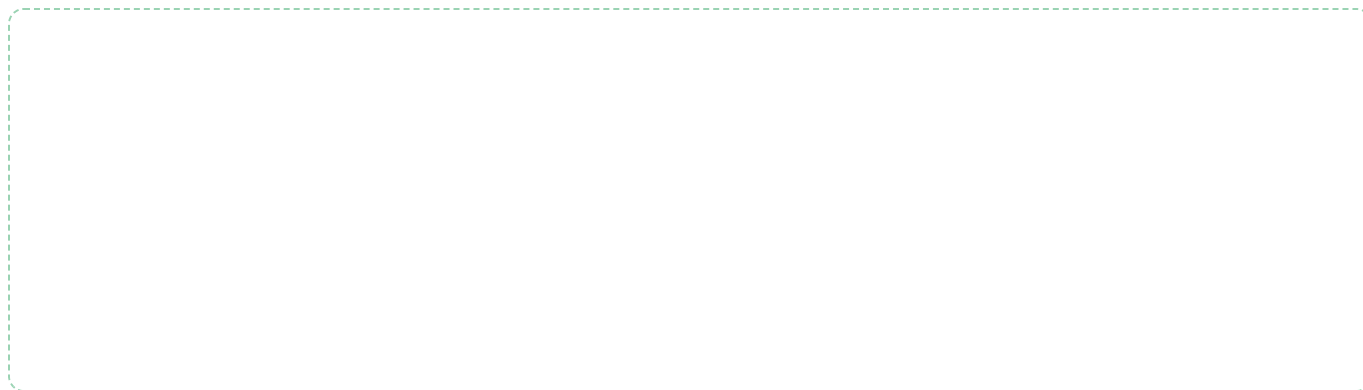
2. Name two things a force can do to a moving object.

3. Which force pulls objects down toward the ground?

4. Which force slows things down and is greater on rough surfaces?

Round 2 — Sketch It

Draw a soccer ball being kicked. Use an arrow to show the direction of the push force. Draw a second arrow to show which way the ball moves.



Force & Motion — Apply It

Keep the cue card covered. Answer first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Idea

1. You push a shopping cart to make it move, then pull it back toward you to stop it. Is each action a push or a pull?

2. A ball rolls faster on a smooth floor than on a rough carpet. Which force explains the difference, and on which surface is it greater?

3. You drop a pencil and it falls straight to the floor. Which force pulled it down, and in what direction does that force pull?

Round 4 — Reason from Evidence

Two students roll the same toy car down two ramps. On the smooth ramp the car travels 3 meters. On the sandpaper ramp it travels only 1 meter. Use the idea of friction to explain why the car went a shorter distance on the sandpaper.

Force & Motion — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain how a force can change the motion of an object. Give two different examples of a change it can cause.

C**Connect**

Connect friction to something you do to control it, such as wearing rubber-soled shoes or oiling a squeaky door. Explain how it helps.

E**Extend**

Plan how you would measure the motion of a marble rolling across a table. Tell what two measurements you would take.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Force & Motion

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. A force is a push or a pull on an object.
2. Any two: start it, stop it, speed it up, slow it down, or change its direction.
3. Gravity.
4. Friction.

Sketch It — look for

An arrow showing the push (kick) force on the ball and a second arrow showing the ball moving in the same direction as the push.

Apply It

1. Pushing the cart to move it is a push; pulling it back to stop it is a pull.
2. Friction; it is greater on the rough carpet, which slows the ball.
3. Gravity pulled it down; gravity pulls objects down toward the ground (toward the center of Earth).

Reason from Evidence — look for

The sandpaper is a rough surface, so it makes more friction. More friction pushes against the car's motion and slows it down, so the car travels a shorter distance.

ACE — Extend look-for

Plan measures the distance the marble travels and the time it takes, since motion is described using distance and time.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color.